Zen — User Guide

兄弟，2026年了，你是不是还在把长长的case列表放在一个device上“玩贪吃蛇游戏”？那Zen可能会对你有所帮助，仅此向MapReduce和StreamJoining致敬

Zen is a dark-themed web console for Roku QA / test-automation. From one UI you can manage device configurations, build and run test profiles, run individual Roku tests locally, browse and cache TestRail results, run AI root-cause analysis on failed logs (and keep a history), talk to an AI assistant with access to Roku internal data (JIRA / Confluence / TestRail / GitLab), transfer files and run shell commands / interactive SSH sessions, keep a library of reference links and docs, and — as an admin — configure the whole system.

Live instance: http://10.64.16.16:9898/ | **Backend API:** proxied under `/api` to the FastAPI service on port 9888.

> **Default admin account:** `admin` / `Zaq1@wsx`. Use **Admin Login** (top-right) to unlock the **Settings** tab (§11). Without admin login, Settings is hidden and the app runs in normal (non-admin) mode.

1. Getting Started

Open http://10.64.16.16:9898/ in a browser (Chrome/Edge recommended — see the note on Save/Copy below). No login is required for normal use; admin login is only needed for Settings.

1.1 Layout

- **Header bar** (sticky at top) — the brand **Zen**, a theme switcher, and the **Admin Login / Admin Logout** button.
- **Left navigation (Sider)** (sticky) — the tab menu; it scrolls internally if long. The **Settings** entry (admin-only) and the **Collapse** toggle stay pinned at the bottom of the viewport.
- **Content area** — a rounded card showing the current page; this is the part that scrolls.

1.2 Navigation tabs

Default order below. Order and visibility are configurable in **Settings → Tab Settings** (§11.1). **Settings** itself only appears after Admin Login.

Tab	Purpose

Device	Manage device configuration files (the TVs / Roku devices under test).
Profile	Manage test profiles and generate them from cached TestRail data.
Run Case	Run test profiles; watch local runner processes and QA Hub suites; review, export, and AI-analyze results.
Run Case in local	Run a single Roku test locally with a live log tail; browse, filter, and AI-analyze past logs.
Analyze History	Browse, filter, view, and delete the AI root-cause analyses saved from Run Case / Run Case in local.
TestRail Case	Manage TestRail plans and search their cases by status (results are cached).
Prompt	AI assistant (RAG) over Roku internal data, with multi-turn chat sessions and reusable prompt Skills.
Reference	A library of reference links and uploaded Markdown docs.
File Transfer	Browse/upload/transfer files, serve them over HTTP, and run shell commands / interactive SSH sessions.
Settings *(admin only)*	Configure tab visibility/order, agent settings, and account credentials/paths.

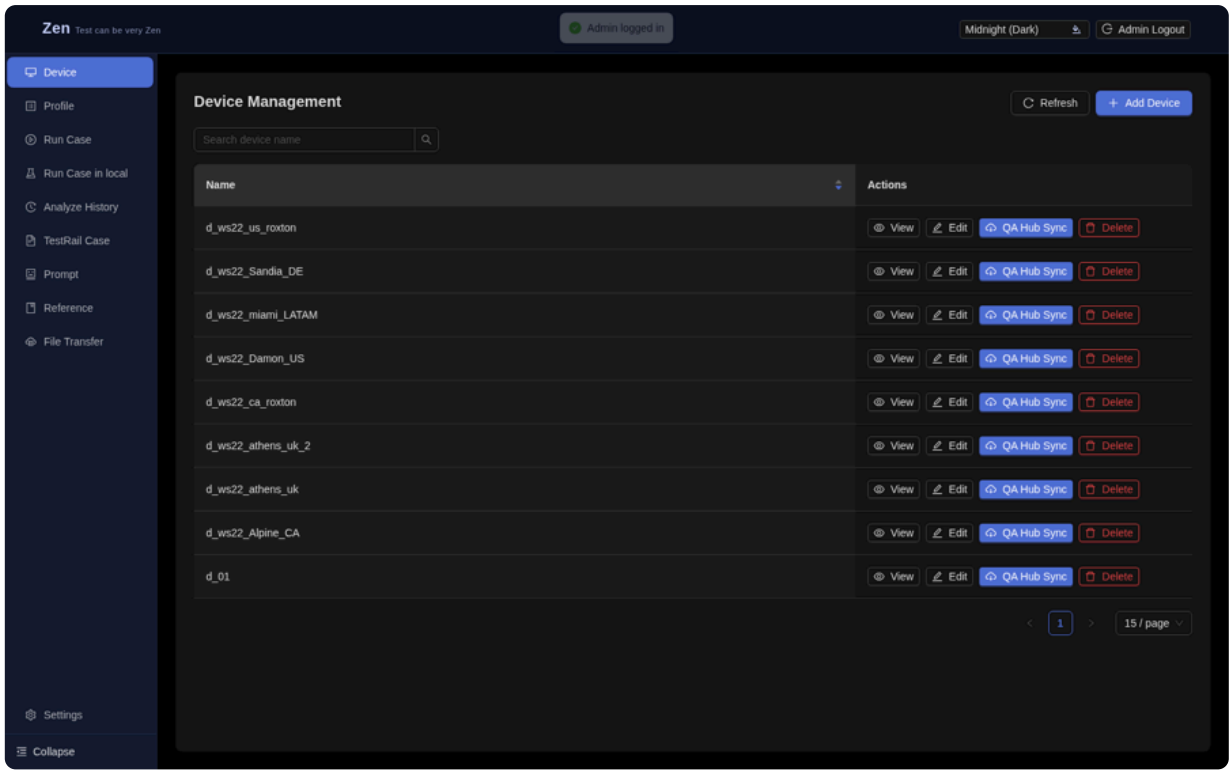
1.3 Theme

Use the header theme switcher to pick from 19 themes grouped into **Dark** (10 — Midnight is the default) and **Light** (9). Your choice is remembered in the browser.

1.4 Admin Login

Click **Admin Login** (top-right), enter the admin username and password (default `admin` / `Zaq1@wsx`), and submit. On success the **Settings** tab appears in the sider and the button becomes **Admin Logout**. The admin state lives in the browser session only (a page reload returns to non-admin). Credentials are verified server-side (bcrypt); the password is never stored or echoed.

2. Device Tab



Lists every device configuration file. Each device is a JSON file describing a TV/Roku under test (IP, brand, power strip, QA bot, serial console, etc.).

2.1 Browsing devices

The table shows one **Name** per device (sortable). Use **Search** to filter; paging at the bottom; **Refresh** reloads from the server.

2.2 Viewing / editing a device

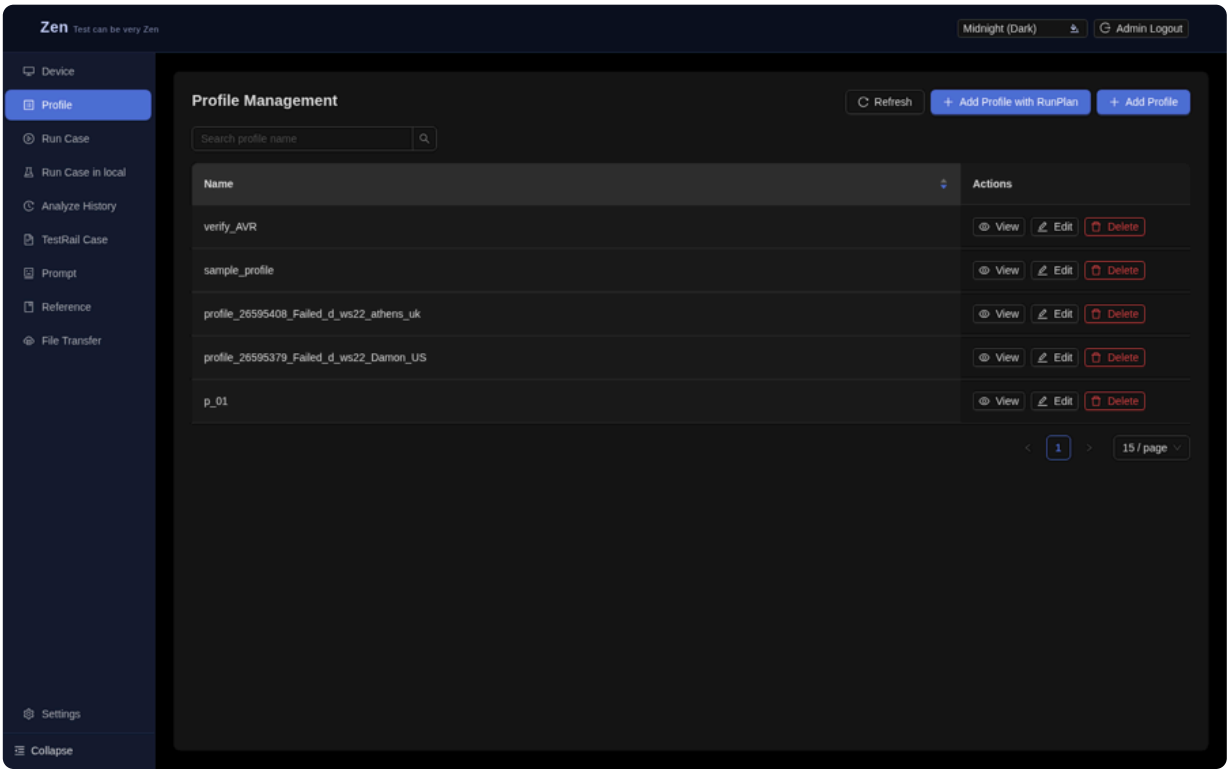
- **View** — read-only JSON with syntax highlighting.
- **Edit** — editable JSON editor; validated as JSON before **Save**.
- **Sync** — refresh the device config from QA Hub.

- **Delete** — remove after confirmation.

2.3 Adding a device

Click **Add Device**, enter a **Name** (without `.json`; becomes the device key used in profiles, e.g. `d_ws22_Sandia_DE`), fill the **Content** (JSON — needs at least `ip_address`, `brand_name` and the ids under a `roku` object), and **Save**.

3. Profile Tab



A profile is a JSON file describing what to run: which cases, on which devices, in what order.

3.1 Managing profiles

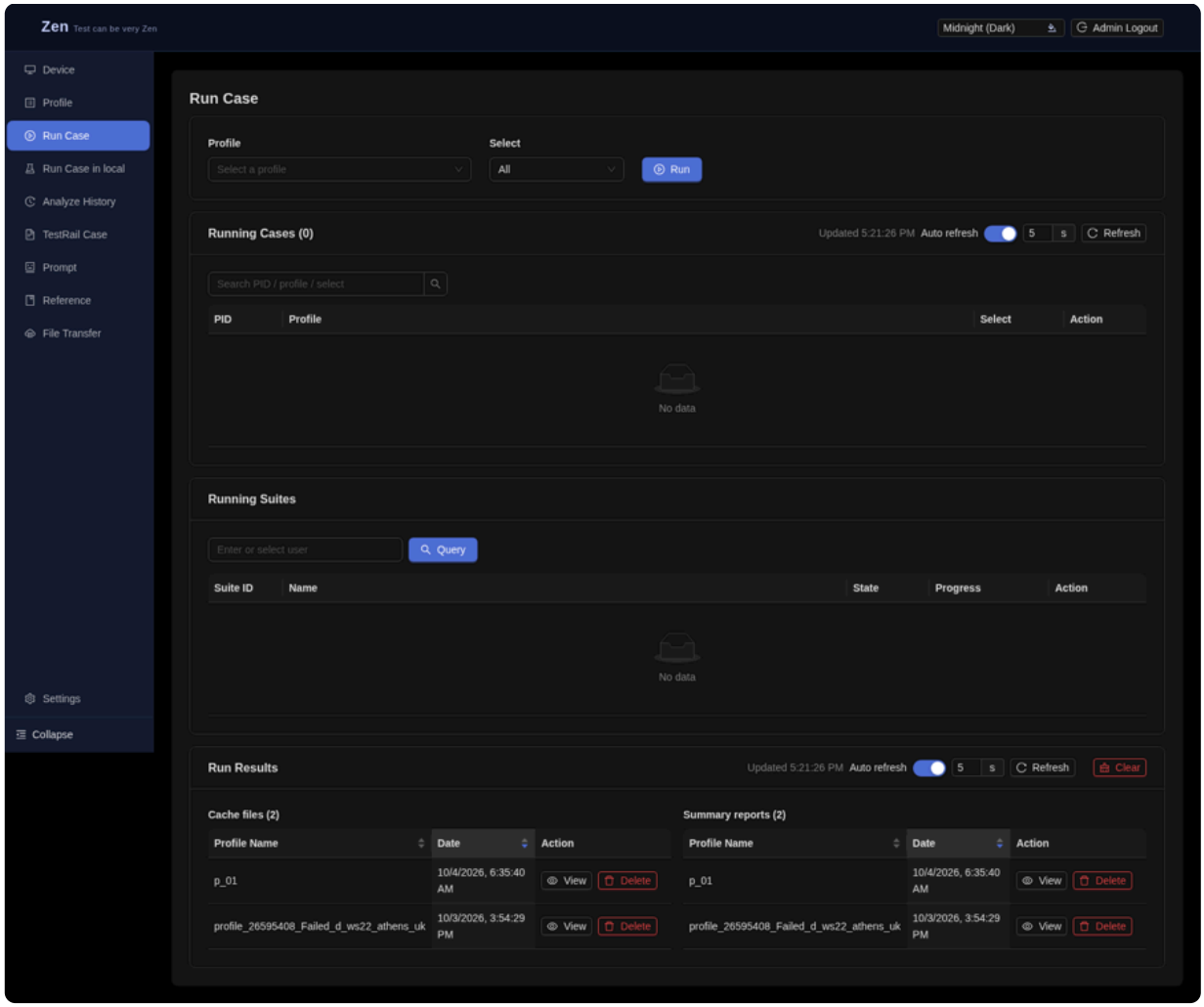
Browse / Search / View / Edit / Delete / Add work like the Device tab (same JSON editor).

3.2 Add Profile with RunPlan (generate from TestRail)

1. Click **Add Profile with RunPlan**.
2. **Plan** — only plans with cached TestRail data appear (shown as `id - name`).
3. **Status** — only statuses cached for that plan (e.g. Failed); disabled until a plan is picked.
4. **Device** — the device to run on.
5. **Add** — builds `profile_<plan>_<status>_<device>.json` (one task per case; each case's automation id becomes a BDD/UT config).

If a plan/status is missing, cache it first in the TestRail Case tab (§7.2).

4. Run Case Tab



Launches test profiles and gives three live views: local runner processes, QA Hub suites, and result files.

4.1 Running a profile

Pick a **Profile** and a **Select** subset (All, Fail, Pass, Skipped, Blocked, NA, Unknown, NoTest), then **Run**. A banner shows the message and PID.

4.2 Running Cases (local processes)

Lists running `test_runner.py` processes (PID, Profile, Select, Stop). Toggle **Auto refresh**, set the interval, or **Refresh**.

4.3 Running Suites (QA Hub)

Pick/enter a user, **Query**; the table lists that user's running QA Hub suites: Suite ID, Name (links to the Jenkins build console), State, Progress, Terminate.

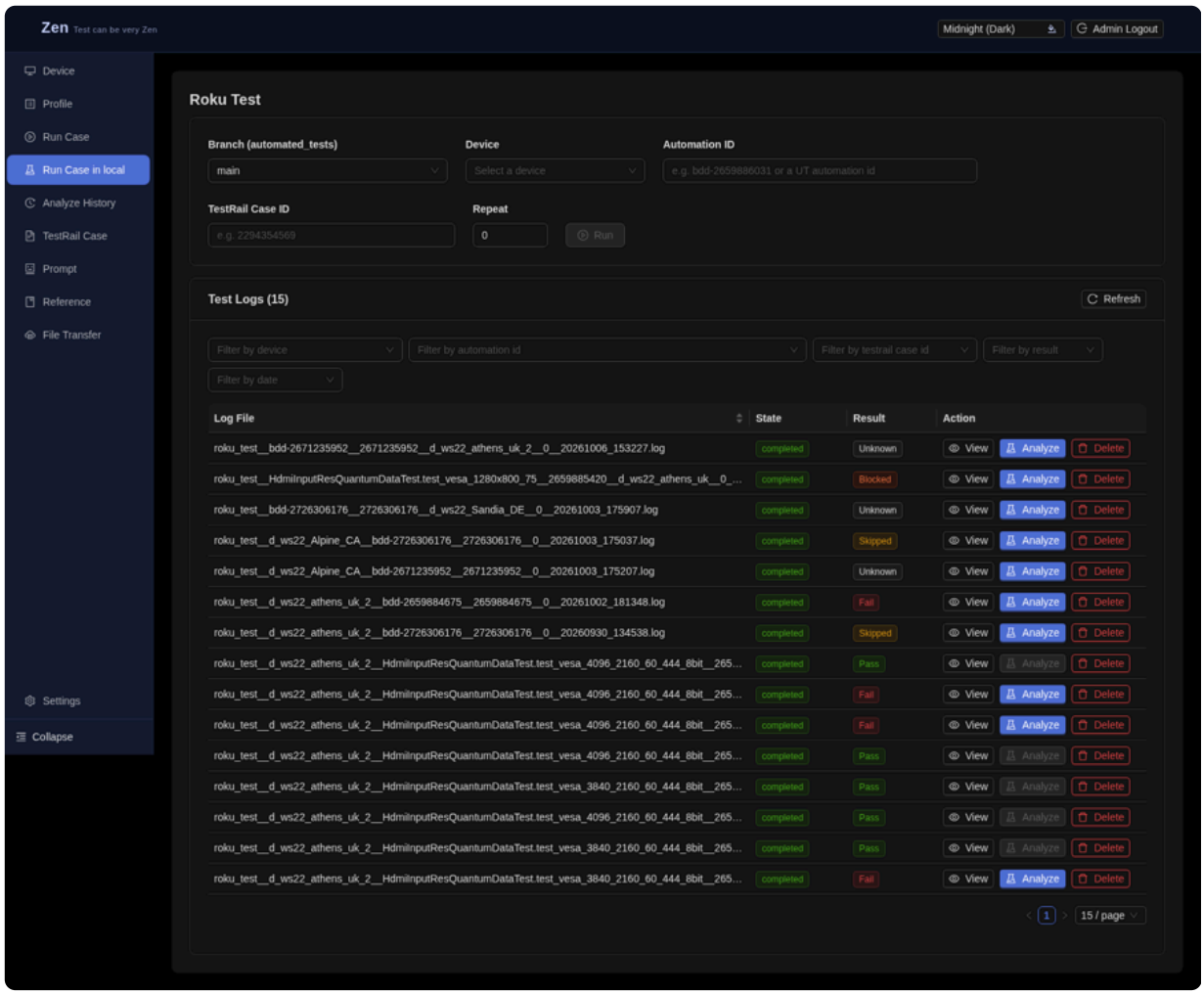
4.4 Run Results (cache & reports)

Two tables (Cache files, Summary reports), each with Profile Name + Date and adjustable refresh. **View** opens a per-run detail table (TestRail Case ID, Automation ID, Status, Build, Log, and an **Analyze** action). **Export Markdown** downloads the table; **Delete** removes a file; **Clear** removes all.

4.5 Analyze a result

Click **Analyze** on a result row to run AI root-cause analysis: Zen fetches that case's Jenkins log server-side and sends it with the configured Analyze prompt template (§11.2). The analysis opens in a Markdown dialog (switch language en/zh to re-run) and is saved to **Analyze History** (§6). Default language and template are set in Settings → Agent Settings.

5. Run Case in local Tab



Runs a single Roku test directly (via the `roku_test` CLI) and tails its log live.

5.1 Running a test

- **Branch (automated_tests)** — pick a git branch; selecting it checks it out on the server.
- **Device**, **Automation ID** (e.g. `bdd-2659886031`), **TestRail Case ID**, **Repeat**.

Run is enabled only when Device, Automation ID and TestRail Case ID are all filled. After starting, a banner shows the PID, live status, and the exact command.

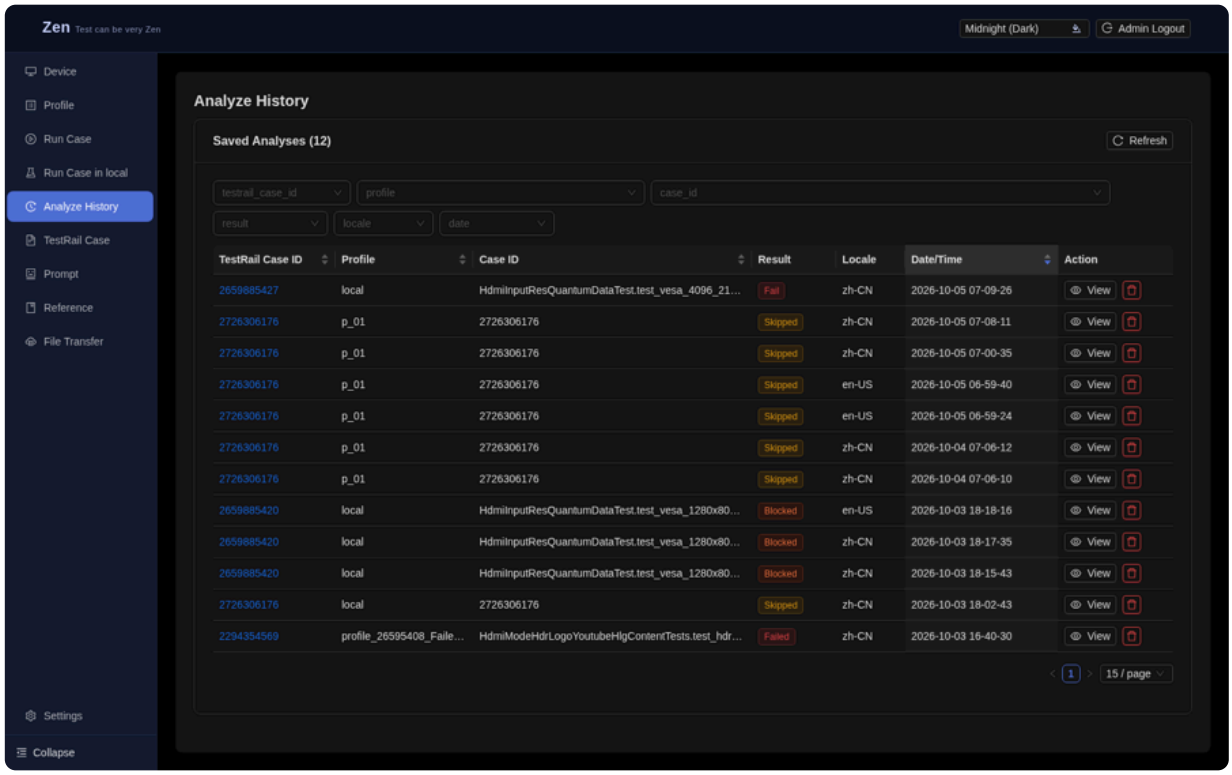
5.2 Live log

Streams the running test output (error lines red, warnings amber), auto-scrolling; survives a page refresh while the test runs.

5.3 Test Logs

Past logs with filters (device, automation id, testrail case id, result, date). Columns: Log File, State, Result (color-coded), Action. **View** opens the log; **Analyze** runs AI analysis (disabled when Pass; tagged with profile `local` in Analyze History); **Delete** removes it.

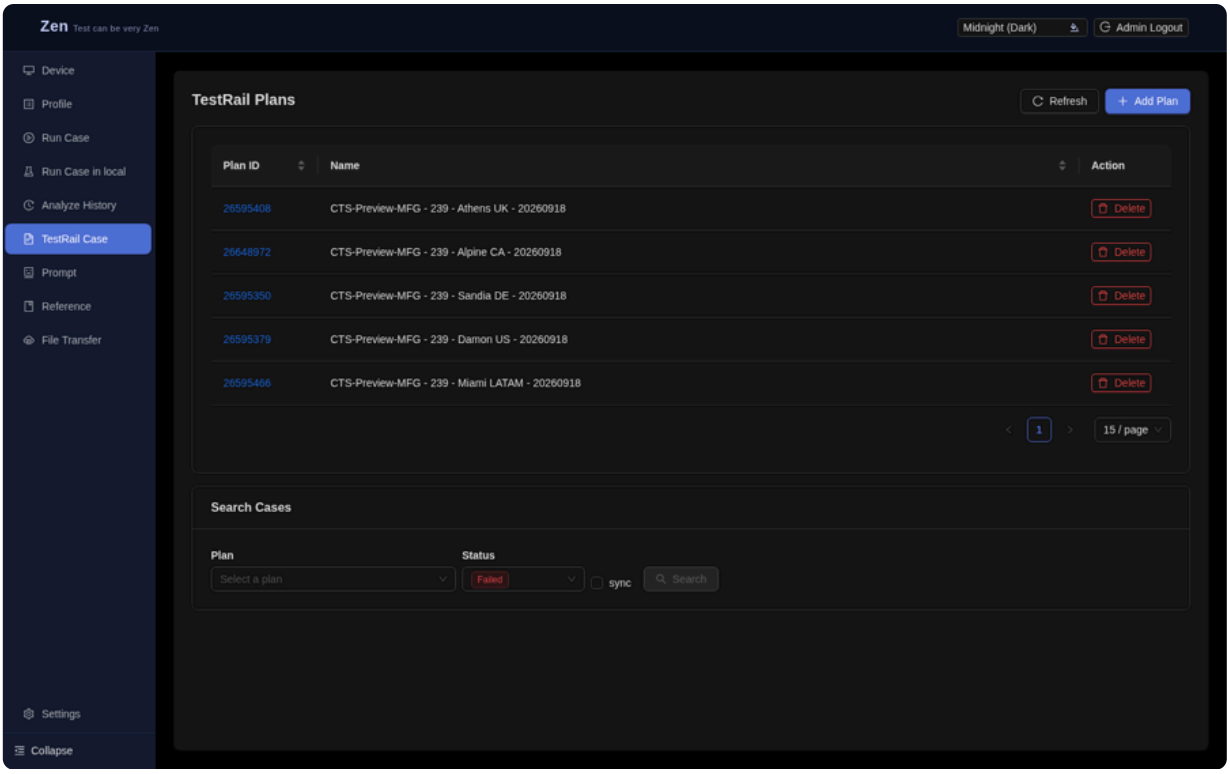
6. Analyze History Tab



Every AI analysis from Run Case (§4.5) or Run Case in local (§5.3) is saved here as Markdown.

- **Filters**: TestRail Case ID, Profile, Case ID, Result, Locale, Date.
- **Columns**: TestRail Case ID (links to TestRail when numeric), Profile, Case ID, Result, Locale, Date/Time.
- **Actions**: **View** (renders Markdown), **Delete**.

7. TestRail Case Tab



Manage TestRail plans and search their cases by status; results are cached to feed the profile generator.

7.1 Plans

The Plans table lists saved plans (Plan ID links to TestRail, Name, Delete). **Add Plan** with the Plan ID (e.g. `26595408`) and a Name.

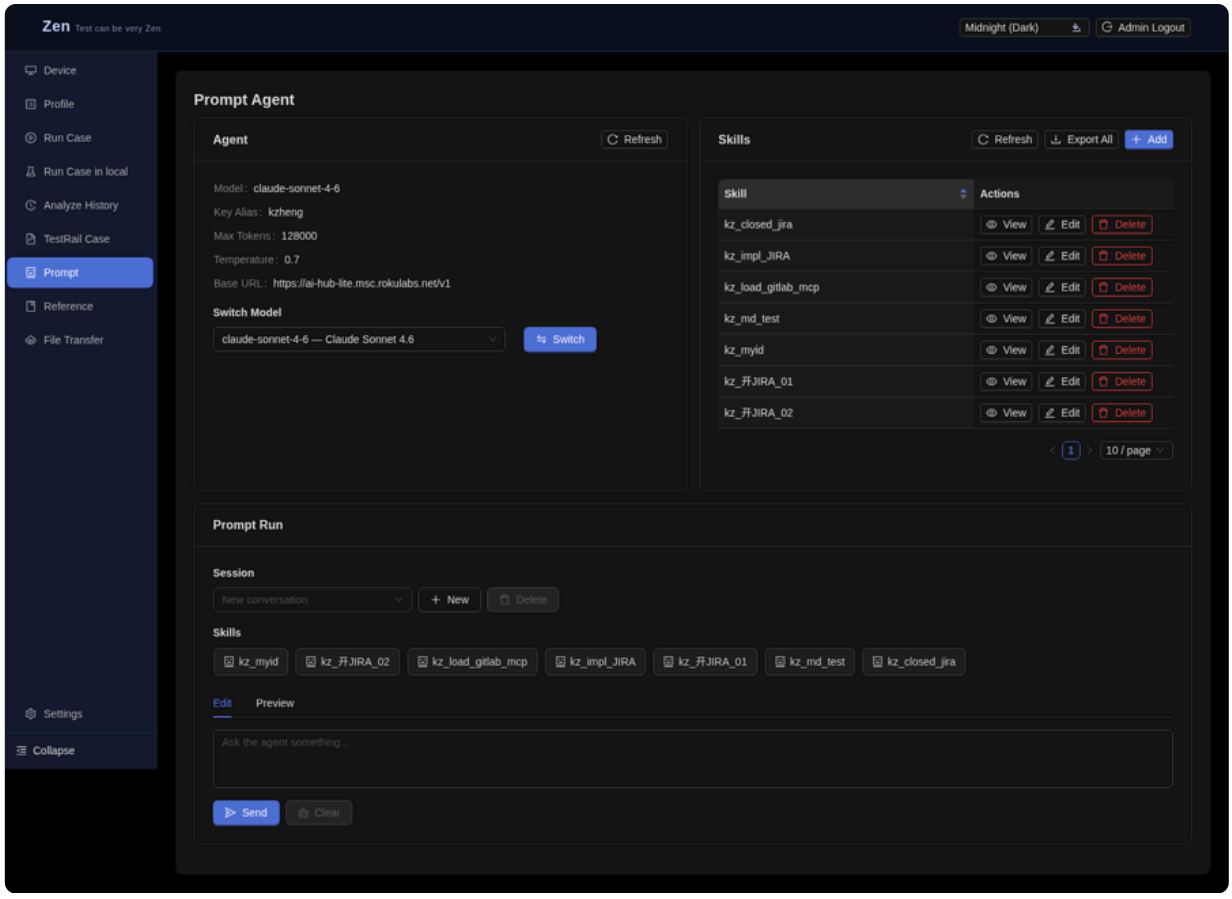
7.2 Searching cases

In **Search Cases**, pick a **Plan** and **Status** (colored tags), optionally tick **sync** to bypass the cache, then **Search**. Columns: Case ID (links to TestRail), Title, Automation ID, Status, Run ID.

7.3 Feeding the profile generator

A search caches results as `<plan>_<status>.json`; that plan+status then becomes selectable in **Profile → Add Profile with RunPlan** (§3.2).

8. Prompt Tab (AI Assistant)



An AI assistant that answers using Roku internal data (JIRA, Confluence, TestRail, GitLab) via retrieval-augmented generation (RAG), with multi-turn sessions and reusable Skills.

8.1 Agent card — model info & switching

Shows the active Model, Key Alias, Max Tokens, Temperature, Base URL. Pick a model from **Switch Model** and **Switch**. The model list is managed in Settings (§11.2).

8.2 Sessions (multi-turn chat)

The **Session** bar: **Select** a saved session (loads its history), **New**, or **Delete**. Each session keeps its messages so the assistant has context; the title comes from the first prompt.

8.3 Skills

A **Skill** is a saved Markdown fragment. The Skills card lists them with **View**, **Edit**, **Delete**, **Add**, and **Export All** (downloads the whole skill library as a zip). Each skill also appears as a clickable **chip** in Prompt Run to append its content.

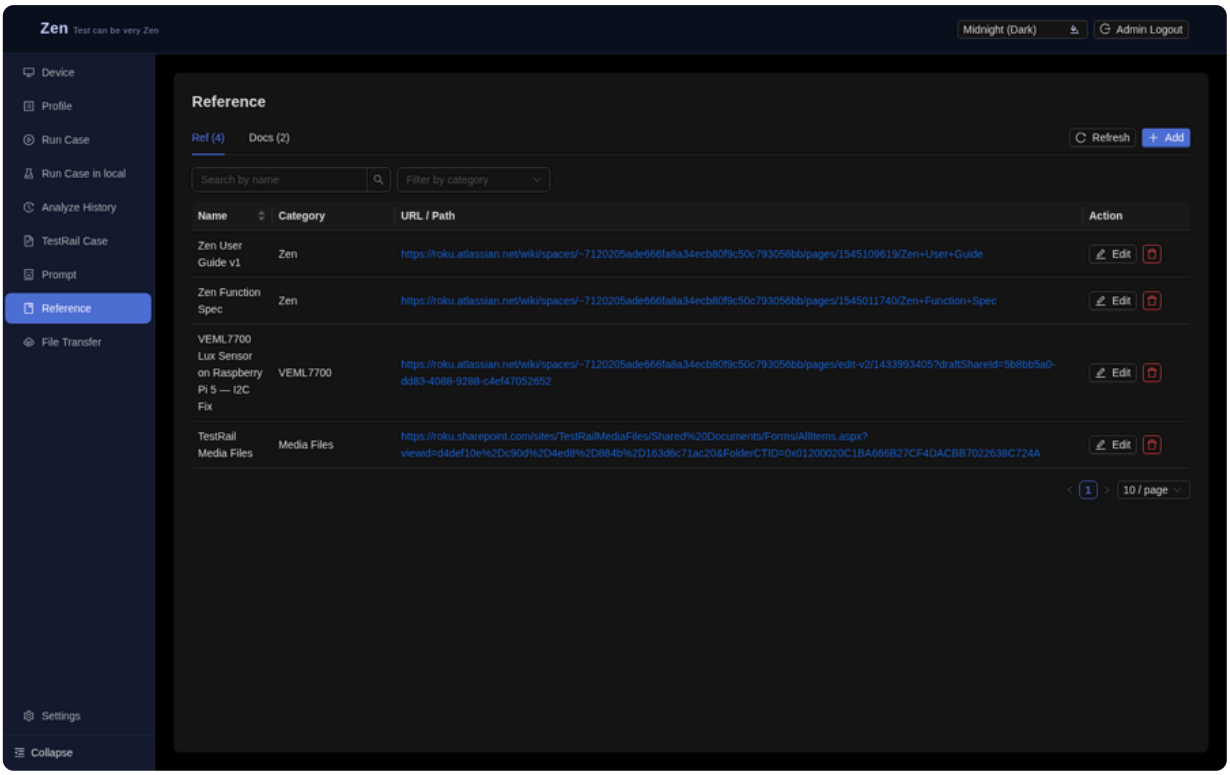
8.4 Prompt Run

Type your question (Edit/Preview Markdown tabs; click skill chips to prepend fragments) and **Send**. Zen runs with internal-data access (JQL path then semantic fallback). **Clear** resets; the answer renders as Markdown with **Save** (.md) and **Copy**.

8.5 Search card

Direct semantic search over indexed internal data: enter a query, pick **Sources** (JIRA, Confluence, TestRail, GitLab, File, S3, Web), **Search**; results list the Document (linked) and a preview.

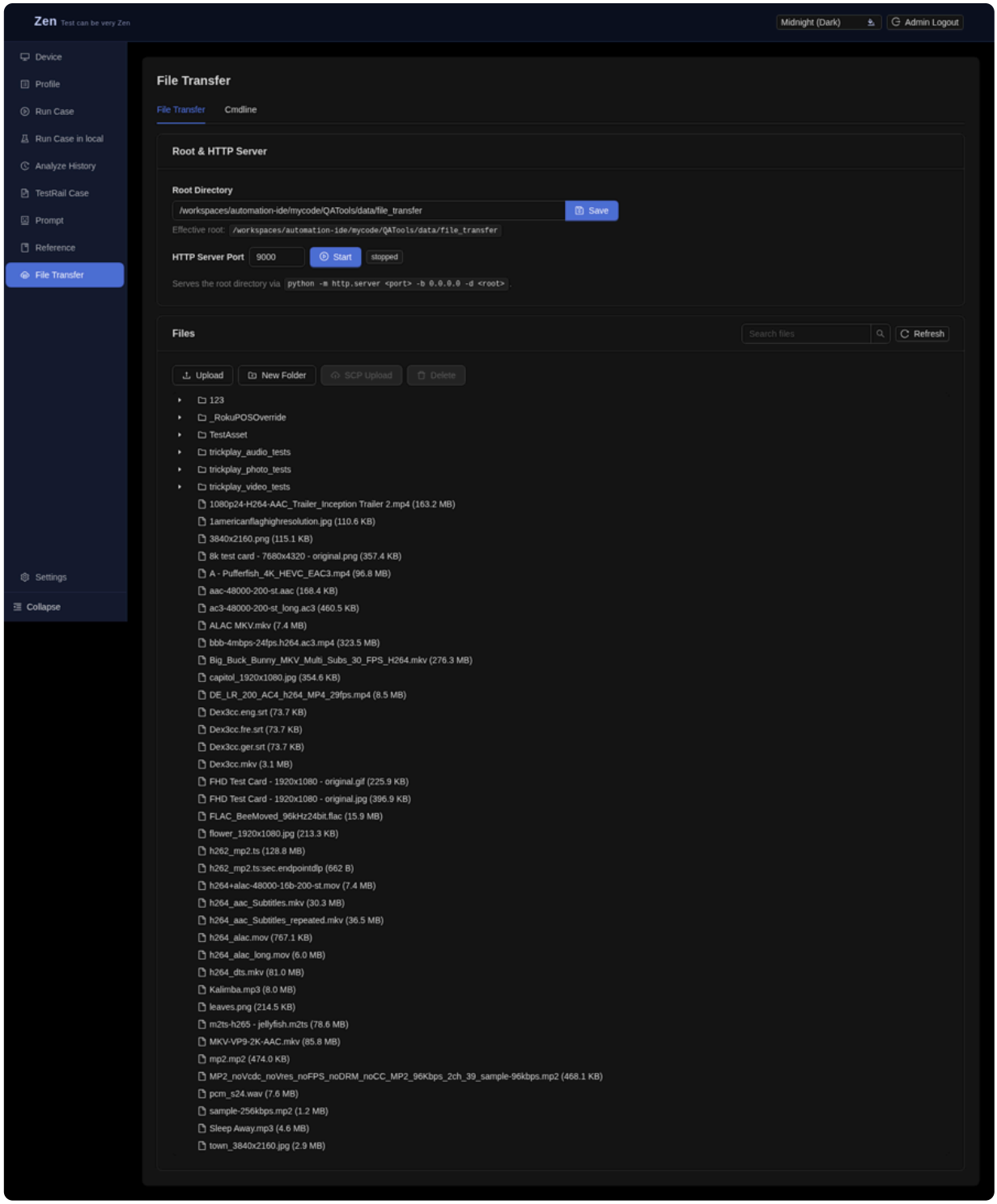
9. Reference Tab



A small team library with two sub-tabs:

- **Ref** — reference links; each entry has Name, Category, URL (clickable). Add/Edit/Delete; filter by name or category.
- **Docs** — uploaded Markdown docs; **Add** accepts a `.md` file (same name overwrites), **View** renders it. Deleting a doc record keeps the underlying file.

10. File Transfer Tab



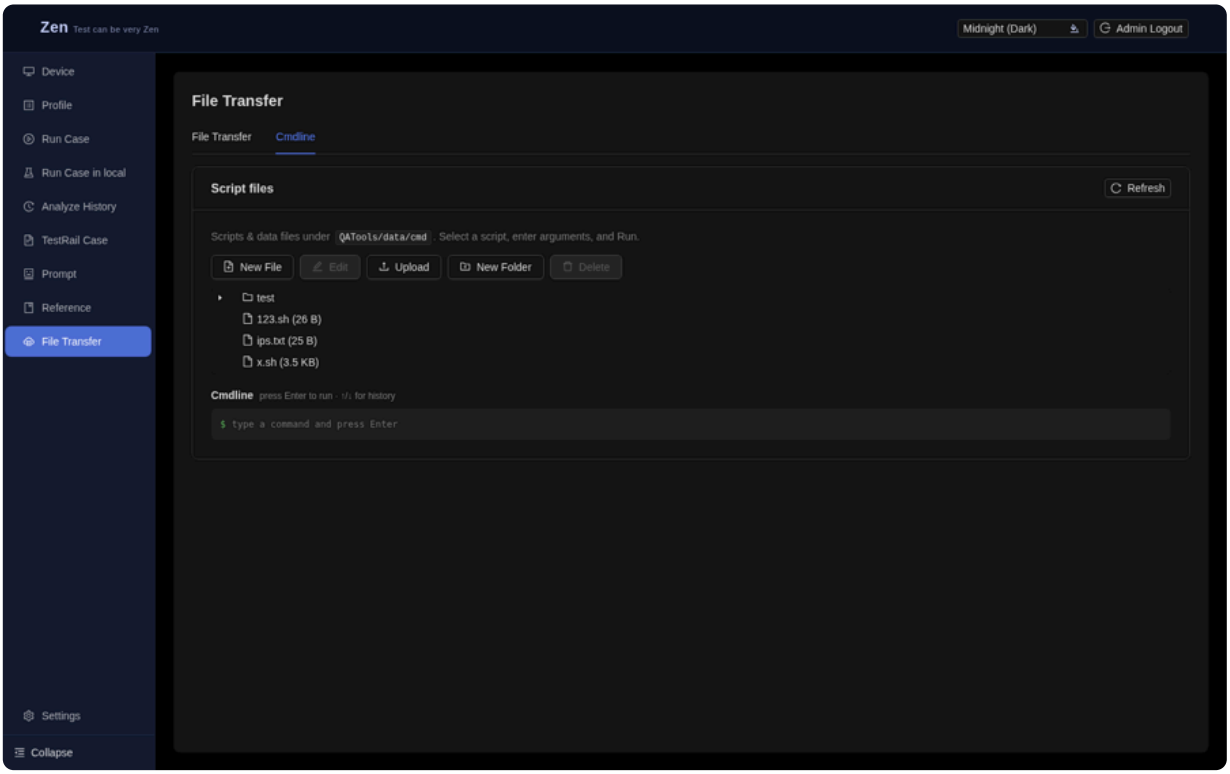
Two sub-tabs: **File Transfer** and **Cmdline**.

10.1 File Transfer sub-tab

- **Root & HTTP Server** — edit the managed **Root Directory** (default `QATools/data/file_transfer`) and **Save**; set the **HTTP Server Port** (default 9000) and **Start** / **Stop** a `python -m http.server` that serves the root; when running, a live clickable URL is shown.

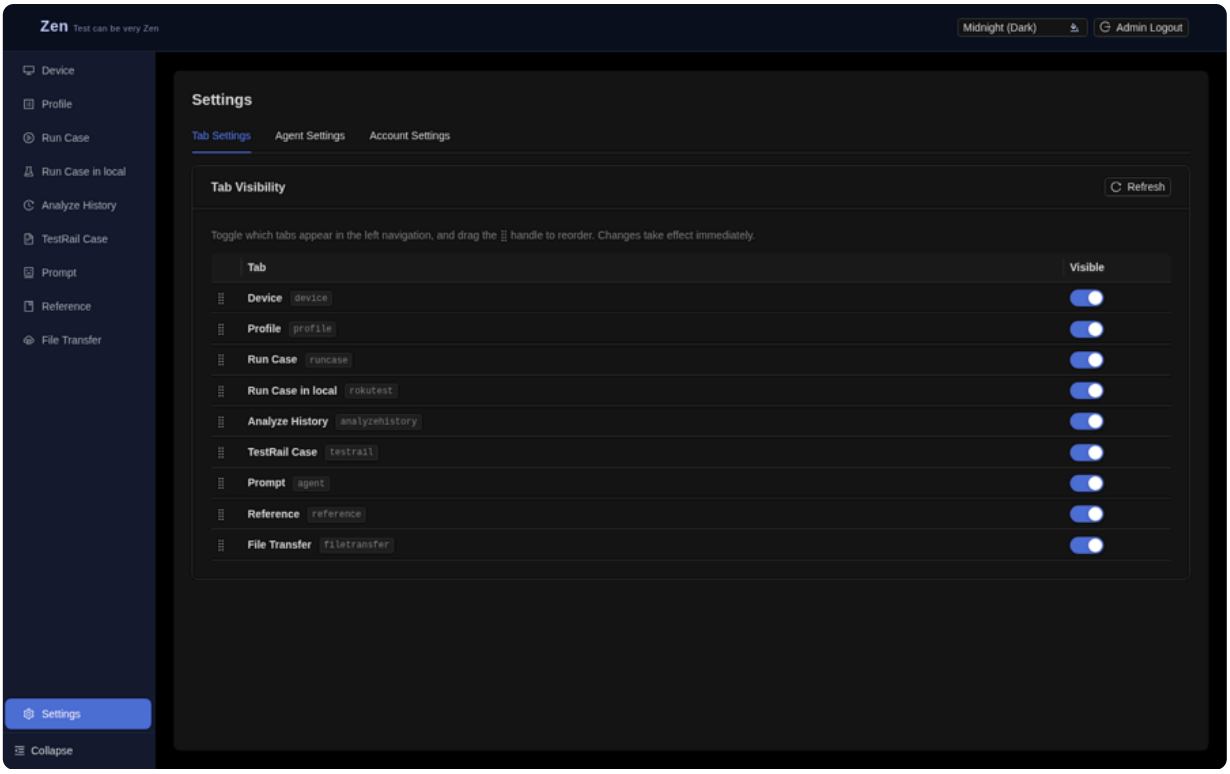
- **Files** — a file tree with **Search files**, **Upload** (with a real progress bar), **New Folder**, **Delete**, and **SCP Upload**.
- **SCP Upload** — a modal where **Host** is an autocomplete of previously-used servers (picking one autofills Remote Directory + User; each remembered host can be deleted). Enter Password; a live progress bar tracks the transfer. Successful targets are remembered.

10.2 Cmdline sub-tab



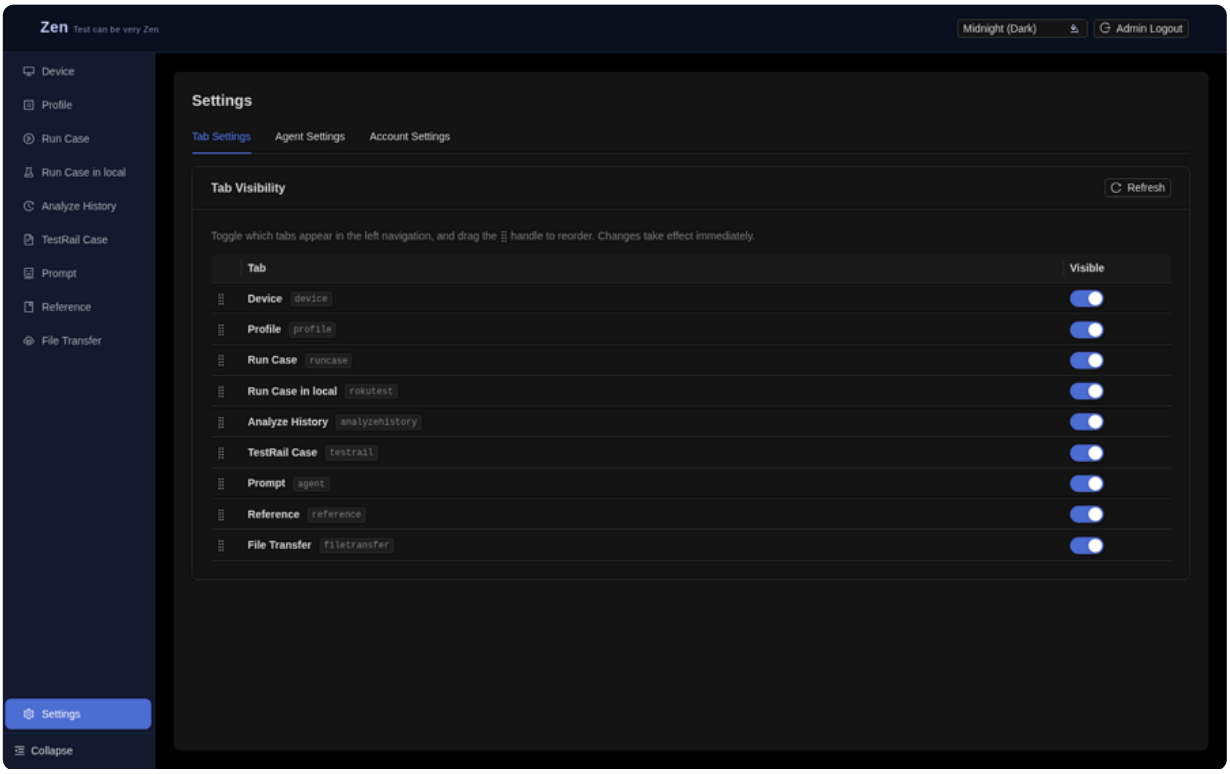
- **Script files** — a tree of scripts/data under `QATools/data/cmd`: **New File** (opens a shell-highlighted editor), **Edit**, **Upload**, **New Folder**, **Delete**. Clicking a file inserts its path into the command line.
- **Terminal console** — type a command and press **Enter** to run; output appears inline (stderr in red, with an `[exit N]` footer). Empty Enter adds a blank prompt line; **↑/↓** recall history; the working directory persists across commands (so `cd ..` works). `ssh` / `scp` with a remote command prompt for a password; a bare `ssh user@host` opens a **persistent interactive SSH session** (the prompt turns amber with `user@host $`; type `exit` to disconnect).

11. Settings Tab (admin only)



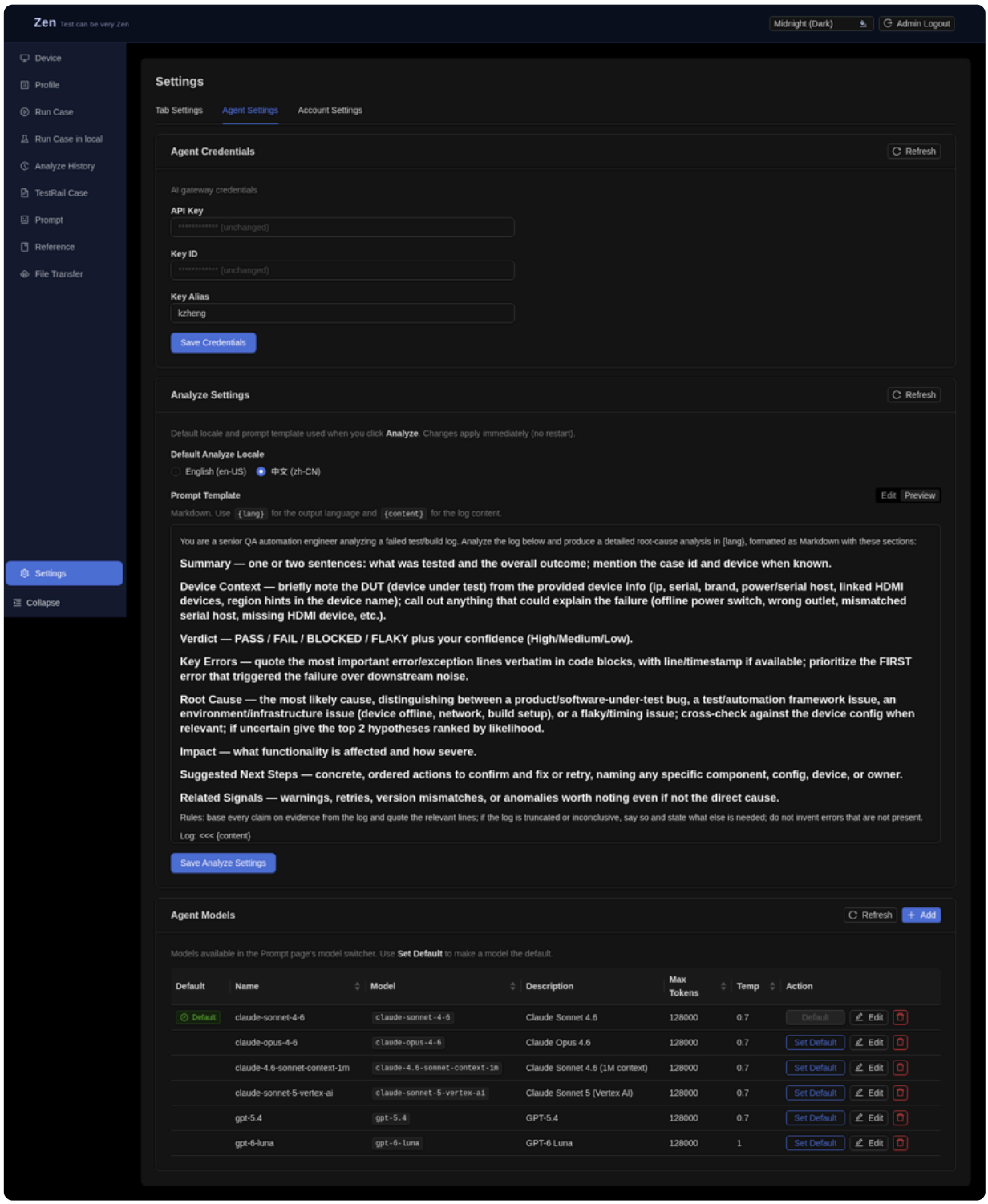
The **Settings** tab is only visible after **Admin Login** (§1.4). It is where the whole system is configured, across three sub-tabs. Some changes apply live; credential/account changes require a backend restart (Zen offers to restart for you).

11.1 Tab Settings



Tab Visibility — a table of all tabs with a drag handle (⠿), the tab label + key, and a **Visible** switch. Toggle to show/hide a tab, or drag rows to reorder them; both apply to the left navigation immediately. (Settings itself is always admin-gated and not in this list.)

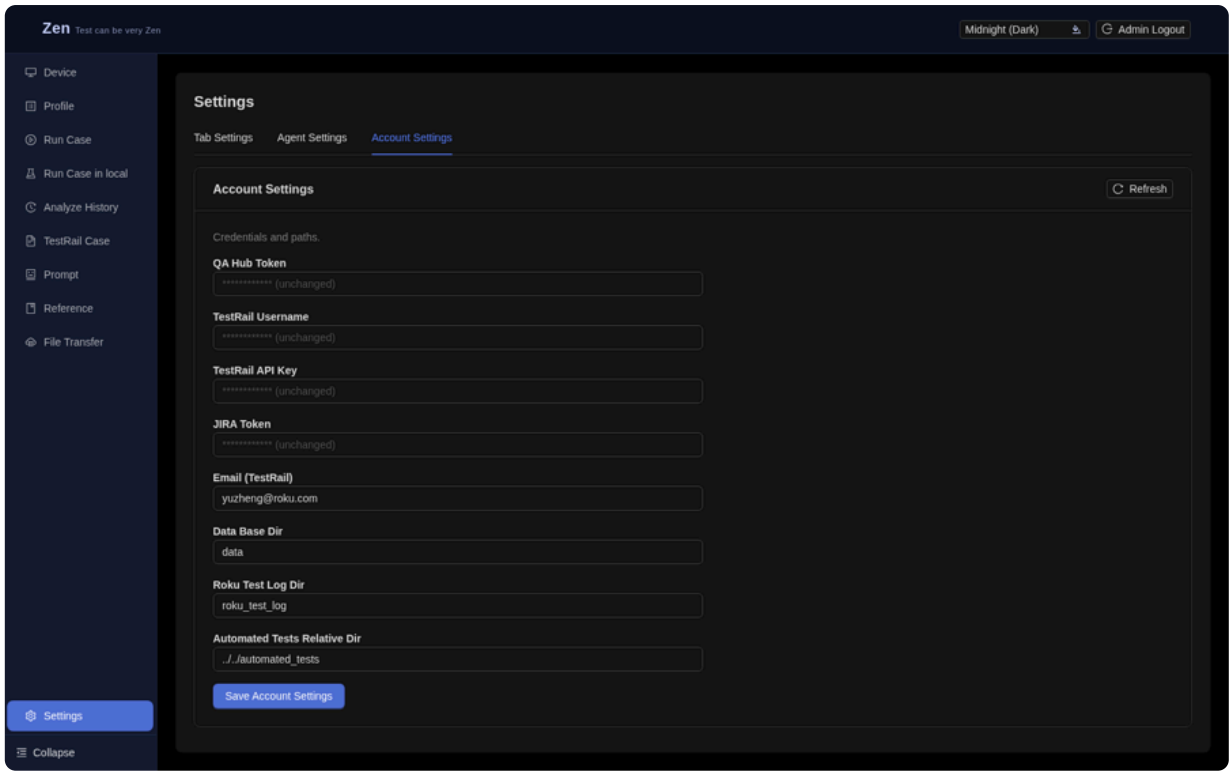
11.2 Agent Settings



Controls the AI used by the Prompt tab and all Analyze actions:

- **Agent Credentials** — the AI gateway auth: **API Key** and **Key ID** (write-only/masked — leave blank to keep the stored value) and **Key Alias** (identifies the caller). Saving offers a backend restart to apply.
- **Analyze Settings** — the **Default Analyze Locale** (English `en-US` / 中文 `zh-CN`) used when you click Analyze, and the **Prompt Template** sent to the AI (Edit / Preview Markdown toggle). The template supports `{lang}` (output language) and `{content}` (the log). Applies immediately — no restart.
- **Agent Models** — the models offered in the Prompt tab's Switch Model dropdown. Add / Edit / Delete a model (Name, Model identifier, Description, Max Tokens, Temperature) and mark one as the **Default** (applied live). The default is the model Zen starts with and uses for Analyze.

11.3 Account Settings



Edits the credentials and paths that live in `config/env.py` (stored as overrides in `config/account.json`). **Secret fields are masked**; leave a secret blank to keep the stored value. **Changes take effect after a backend restart** (Zen offers to restart on save).

Field	Secret	What it controls
QA Hub Token	Yes	Auth for all QA Hub calls (device sync, suite list/terminate).

TestRail Username	Yes	TestRail API username (with the API key) for plan/case fetch.
TestRail API Key	Yes	TestRail API key (Basic auth with Email).
JIRA Token	Yes	JIRA access used by the assistant's internal search.
Email (TestRail)	No	TestRail login email (Basic-auth username).
Data Base Dir	No	Base folder for all runtime data (reports, sessions, analyzer history, skills, docs, file_transfer).
Roku Test Log Dir	No	Where Run-Case-in-local (`roku_test`) logs are written and listed.
Automated Tests Relative Dir	No	Location of the `automated_tests` git repo (branch list/checkout, roku_test target).

12. Tips & Troubleshooting

- **Settings tab missing** — you are not logged in as admin. Click **Admin Login** (default `admin` / `Zaq1@wsx`). A page reload logs you out.
- **Save / Copy** work best in Chrome/Edge (native file picker + clipboard); other browsers fall back to a direct download.
- **"Add Profile with RunPlan" shows no data** — cache that plan+status first in the TestRail Case tab (§7.2).

- **Analyze says the prompt template is not configured** — set it in Settings → Agent Settings → Analyze Settings (§11.2).
- **A tab is missing** — it was hidden in Settings → Tab Settings; re-enable it there.
- **Account/credential change didn't take effect** — those apply after a backend restart; accept the restart prompt on save.